Question 1

Gene expression refers to genes being 'turned on' and producing a product. The product could be an enzyme, a structural protein, or a control molecule.

Eukaryotes can regulate their gene expression at various levels whereas prokaryotes regulate gene expression predominantly at the transcriptional level.

a) Explain how the process of transcription is regulated in eukaryotes. [2]

1. Acetylation of histone increase transcription / deacetylation decrease transcription
2. Methylation of DNA decrease transcription/ Demethylation increase transcription

b) Explain why prokaryotes regulate gene expression predominantly at the transcriptional level. [2]

1. due to absence of a membrane-bound nucleus;
2. which means that transcription and translation occurs simultaneously in prokaryotes; hence by regulating transcription, the bacteria is also regulating translation;

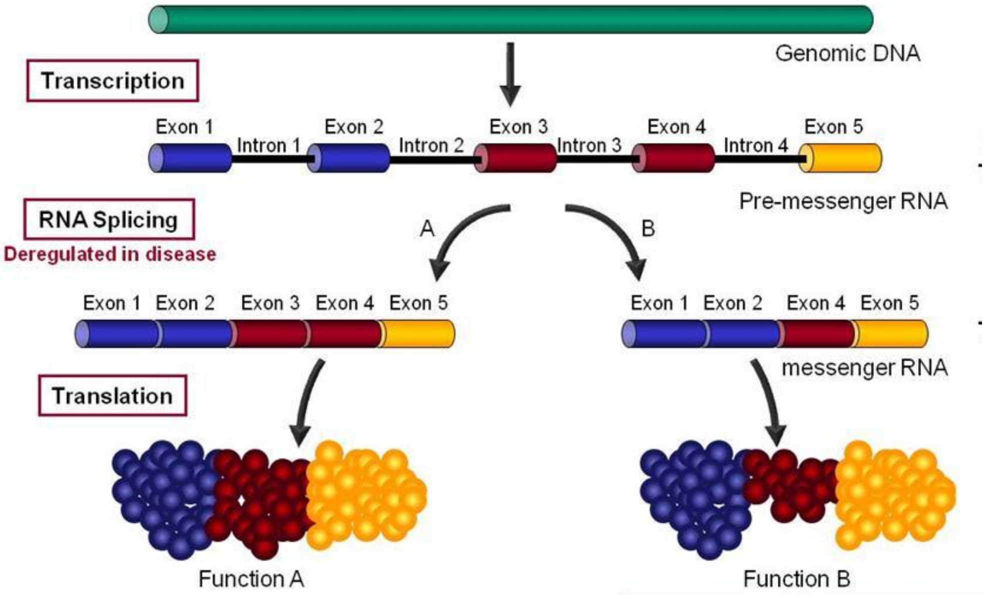
   OR

3. genes are arranged in the form of operons in prokaryotes;
4. transcriptional control allows many genes involved in the same metabolic pathway to be regulated at the same time;

Post transcriptional modification in eukaryotes involve the addition of the 5' guanosine cap, splicing as well as the addition of the 3' polyA tail. Explain what will happen to resulting protein structure if a mutation occurs at the splice site of an intron. [3]

1. Small nuclear ribonucleoproteins (snRNPs) will not be able to recognize and bind to splice site of the intron on the mRNA;

1. spliceosome (pre-mRNA-snRNP complex) is not formed;
2. Thus the intron is not excised out;
3. and the mature mRNA formed is longer than normal;to be translated to a much longer polypeptide than usual;
4. which will not be folded to the same 3D conformation as the normal protein; on functional protein



d) Name and describe the process that is shown in Figure 1.1. [3]

- Alternative splicing
- Spliceosomes recognize the intron- exon junctions and remove all the introns;
- Different pre-mRNA can be spliced into different mature mRNA by the inclusion of different sets of exons
- Resulting in different polypeptides

e) Suggest why there is a need for this process. [1]

- Eukaryote has a larger of non coding region compared to coding region, alternative splicing allows for a single transcript to potentially generate a large number of different protein variants

Question 2

a) Sex hormone estrogen promotes mammary epithelial cell proliferation by stimulating the expression of genes encoding cell cycle regulatory proteins. Estrogen receptors (ERα) are intracellular receptors of estrogen. Estrogen-receptor complexes bind to estrogen response elements (ERE) in enhancers. Estrogenic control of the cell cycle occurs by increasing the expression of *cyclin* and *myc C* which promote cell cycle progression.

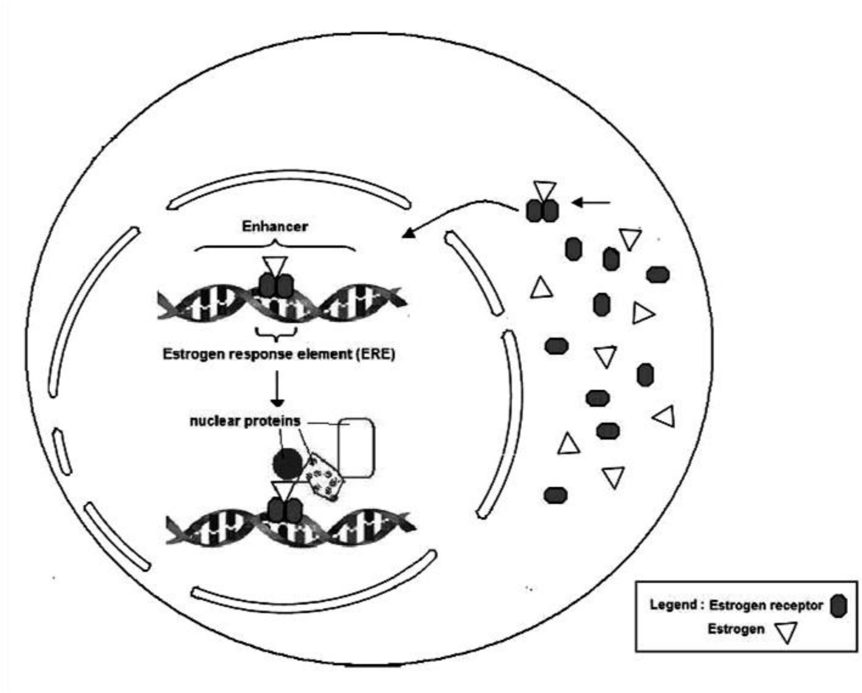


Figure 2.1

With reference to **Figure 2.1**

i) State the level at which gene expression is regulated by estrogen. [1]

transcriptional level;

ii) describe and explain how estrogen regulates gene expression. [4]

- Estrogen lipid soluble, diffuse across phosplipid bilayer to bind to intracellular receptor form hormone receptor complex;

  ER in cytoplasm, estrogen **binds to receptors** and causes dimerisation;

  Conformation change when estrogen bind to receptor => activated

- enters nucleus and binds to ERE on the enhancer sequence,
- Identify hormone receptor complex as an activator
- attracting other general transcription factors and RNA polymerase to bind; (reject nuclear proteins just what exactly are they?
- increases rate of transcription of cyclin and myc Cwhich causes cell division.

b) There are two isoforms of estrogen receptors: ERα and ERβ encoded by different genes. Genomic DNA samples taken from women suffering from breast cancer were analysed for the gene copy number present for the ERα gene.

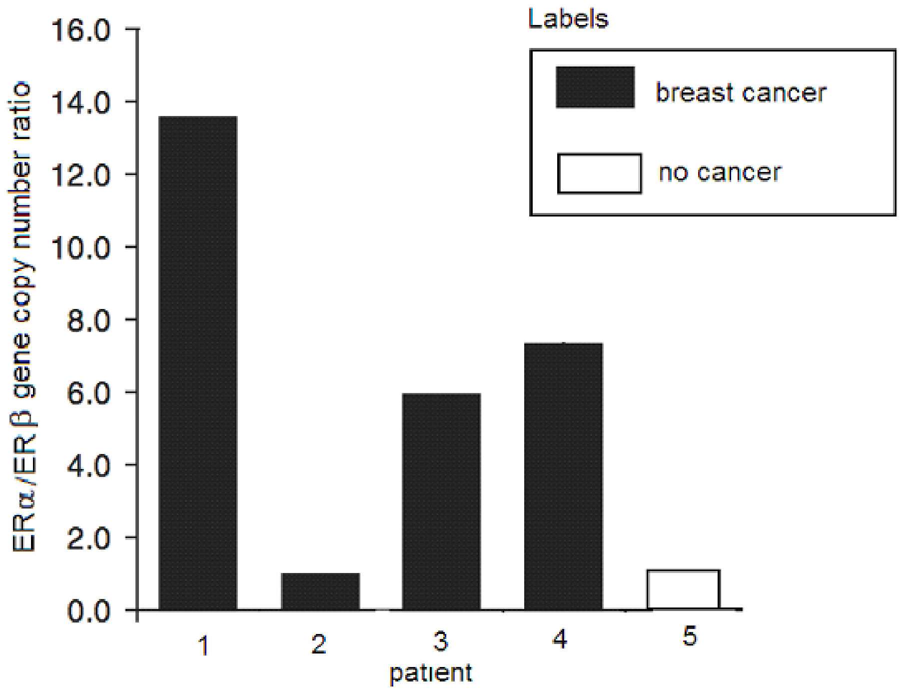


Figure 2.2

i) With reference to the **Figure 2.2**, describe the effect different ERα gene copy number on the women. [2]

- high gene copy number ratio is correlated to occurrence of breast cancer
- Women 1,3, 4 who had 14 copies, 6 copies and 7 copies of ERα gene suffered from breast cancer;
- Woman 2 and 5 had 1 copy of the gene but woman 2 had cancer while woman 5 did not suffer from cancer

c) Suggest an explanation as to why patient **2** developed cancer despite a low copy number. [1]

- G**ain of function mutations** on **other** proto-oncogenes. Just require 1 copy of proto-oncogene to be converted to oncogene
- Loss of function of tumour suppressor genes both copies or alleles NOT both genes
- **This results in** permanent activation of hormone receptor complex

d) Suggest how amplification of the ER gene contributes to breast cancer. [2]

- **more** intracellular **receptors** are synthesized due to increased number of copies;
- **more activators bind** to enhancers to **increase** rate of transcription of genes encoding proteins which trigger cell division/ proto-oncogenes;

[Total: 10 marks]

Question 3

a) A homozygous black mouse was crossed with a homozygous recessive white mouse. All the F1 offspring were black. When the F1 offspring were selfed, the ratio of black to brown to white was 9:3:4.

Represent the above information in the form of a genetic cross. [4]

Parental phenotype	Homo black mouse	x	Homo white mouse
Parental genotype	AABB	x	aabb
Gametes	AB	x	ab
F1 genotype	AaBb		
F1 phenotype	AaBb	x	AaBb
Gametes	AB Ab aB ab		AB Ab aB ab

Punett Square

	AB	Ab	aB	Ab
AB	AABB black	AABb black	AaBB black	AaBb black
Ab	AABb black	AAbb brown	AaBb black	Aabb brown
aB	AaBB black	AaBb black	aaBB white	aaBb white
Ab	AaBb black	Aabb brown	aaBb white	Aabb white

Phenotypic ratio: black: brown: white
9:3:4

b) Explain the molecular mechanism of this cross. [3]

-Recessive epistasis
-2 recessive alleles for gene A will mask the effect of Gene B
-aa_ _ =white
-A_B_=black
-A_bb =brown

c) **Table 3.1** shows the inheritance of kernel colour in wheat when 2 intermediate parents were crossed. (AaBbCc x AaBbCc). The offspring contain seven different shades of kernel color based on the number of capital letters in each genotype. The higher the number of capital letters, the darker the shade of the wheat kernel.

Gametes	ABC	ABc	AbC	Abc	aBC	aBc	abC	abc
ABC	6	5	5	4	5	4	4	3
ABc	5	4	4	3	4	3	3	2
AbC	5	4	4	3	4	3	3	2
Abc	4	3	3	2	3	2	2	1
aBC	5	4	4	3	4	3	3	2
aBc	4	3	3	2	3	2	2	1
abC	4	3	3	2	3	2	2	1
abc	3	2	2	1	2	1	1	0

Table 3.1

i) Identify and state the characteristics of such an inheritance. [3]
-Polygenic inheritance
-Additive effect
-Explain additive:The colour is the combined effect of many different genes on multiple loci.
-Continuous variation/ range of phenotype in F2

Question 4

a) Antibiotic medications are used to kill bacteria, which can cause illness and disease. They have made a major contribution to human health. Many diseases that once killed people can now be treated effectively with antibiotics. However, some bacteria have become resistant to commonly used antibiotics.

Antibiotic resistant bacteria are bacteria that are not controlled or killed by antibiotics. Due to mutation, they are able to survive and even multiply in the presence of an antibiotic. Most infection-causing bacteria can become resistant to at least some antibiotics. Bacteria that are resistant to many antibiotics are known as multi-resistant organisms (MROs).

Antibiotic resistance can cause serious disease and is an important public health problem. It can be prevented by minimising unnecessary prescribing and overprescribing of antibiotics, the correct use of prescribed antibiotics, and good hygiene and infection control.

i) Explain why a mutation acquired by a bacteria will most certainly affect its pheonotype. [2]

-monoploid=1 allele (reject haploid)
-No presence of dominant allele to mask the effect

ii) Explain how natural selection has led to the emergence of such antibiotic resistant bacteria. [4]

-Variation: different variation of bacteria some resistant to antibiotic some not
-Selection pressure: Antibiotic
-Selective advantage: Bacteria that is resistant to antiobiotic
-Bacteria survived to produce offspring by binary fission and pass on their antibiotic resistance gene to the next generation=> Frequency of antibiotic resistance allele increases in population.

b) According to the Red Queen hypothesis, sexual reproduction persists because it enables many species to rapidly evolve new genetic defenses against parasites that attempt to live off them.

Scientists have tested this idea by observing different groups of small fish *Poecilopsis* species (Gila Topminnow) in Mexico.

Gila Topminnow. Photo by Arizona Game and Fish, Non-Game Division

Some populations of the topminnow reproduce sexually, while others practice parthenogenesis. Parthenogenesis occurs when females produce offspring without any male contribution and the female's gametes develop directly into female offspring.

Topminnows are constantly parasitized by ‘black spot disease’ caused by black spot worms that encyst in the skin. Parasitized topminnows rarely survive. The researchers found that identical populations of the asexually reproducing topminnows harbored many more black-spot worms than did those producing sexually, a finding that fit the Red Queen hypothesis: the sexual topminnows could devise new defenses faster by recombination than the asexually producing clones.

However, it was observed that after a drought, the sexually reproducing topminnows were more heavily parasitized than the cloned topminnows. This process of whereby chance events cause the allele frequency to change unpredictably is known as genetic drift.

i) With reference to the example of topminnow Explain the concept of genetic drift and how it can alter the population’s gene pool significantly. [2]

-Genetic drift is the random change in allele frequencies due to chance factors/not due to selection pressure
-Chance factor: Drought
-allele that make them resistant to parasite frequency decrease significantly => eventually the entire population may be homozygous

ii) Topminnows are brought by man to a new place with totally different environmental conditions. Over time, speciation occurs and a new species of fish related to the topminnows is discovered.

Explain how speciation occurred to give rise to the new species of fish. [2]

-Geographical isolation prevent the sub-group from interbreeding with the rest of the population -> leading to genetic isolation -> no gene flow
-Different selection pressure

[Total: 10 marks]

Question 5

Figure 5.1 shows a root tip of the onion plant. The meristematic region contains cells undergoing rapid nuclear division.

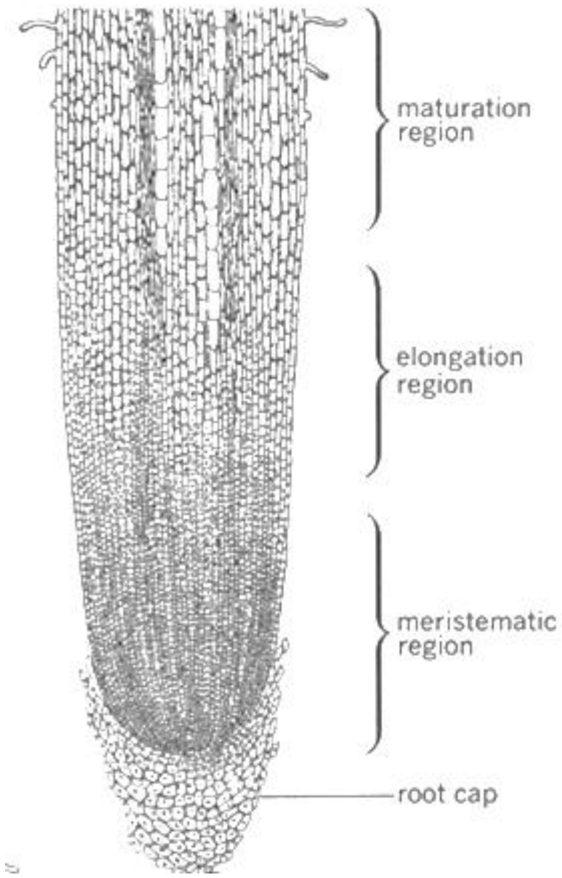


Figure 5.1

(a) Explain the function of the type of nuclear division that is occurring in the cells found in the meristematic region in Figure 5.1. [3]

- Mitosis
- Main function is to maintain genetic stability by maintaining the same number and type of chromosomes during nuclear division
- Thus enabling growth and cell regeneration in the meristamitic region of the root tip.
  (max 3)

(b) Describe how two distinct daughter cells are formed at the end of nuclear division. [2]

- Cytokinesis occurs
- Golgi vesicles containing cellulose move along microtubules to the middle of the cell;
- Membranes of golgi vesicles fuse, cellulose deposited forms cell plate that extends across the equator, forming two distinct daughter cells.

  (max 2)

A researcher randomly extracted two meristematic cells. He proceeded to analyse them and found them to be genetically non-identical. He concluded and wrote down in his research log book that this was due to "independent assortment of chromosomes before non-identical sister chromatids are pulled apart to opposite poles of the dividing cell".

(c) Do you agree with the researcher's conclusion? Explain why. [3]

- Do not agree/ conclusion was Invalid, because mitosis is occurring in meristematic cells instead of meiosis.
- Independent assortment occurs during metaphase I of meiosis, where bivalents/ pairs of homologous chromosomes line up randomly in two rows at the equator.
- However, during metaphase of mitosis, chromosomes with identical sister chromatids align in a single row at the equator thus no independent assortment occurs.

The graph in Figure 5.2 shows the amount of DNA in the nuclei of cells taken from the meristematic region of the onion plant.

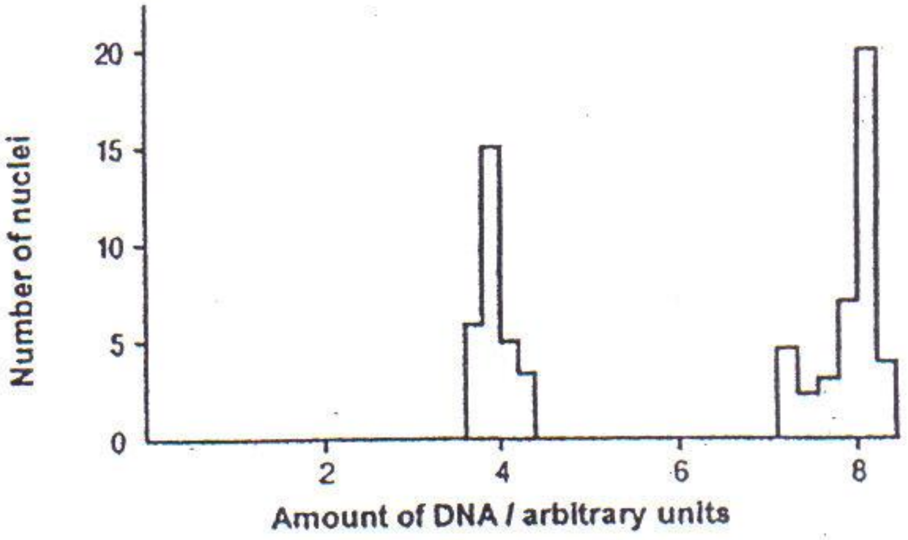


Figure 5.2

(d) With reference to Figure 5.2, state precisely the molecular process that accounts for the change in DNA amount in the nuclei of cells from 4au to 8 au. [1]

- Semi-consvervative DNA replication occurring during S phase of Interphase

(e) Compare the above process in (d) with transcription. [4]

- (similarity) Both DNA replication and transcription has elongation of daughter strands in the 5' → 3' direction.
- (similarity) Both DNA replication and transcription uses DNA as a template to synthesise their respective products.
- (similarity) Both DNA replication and transcription involves complementary base-pairing (A- T/U; C-G with template strand to sequence/ synthesise their respective products.
- (difference) Product of DNA replication is double stranded DNA molecule consisting of one old and one new strand, while product of transcription is a single stranded RNA molecule.
- (difference) DNA replication uses both strands of DNA as a template while transcription only uses antisense strand as template.
- (difference) DNA replication involves the use of thymine to complementary base-pair with adenine, while transcription uses uracil to complementary base-pair with adenine.
  (2 similarities + 2 differences, max 4)

[Total: 13 marks]

Question 6

(a) Figure 6.1 shows a generalized graph of the relationship between HIV copies and CD4 counts over the average course of untreated HIV infection.

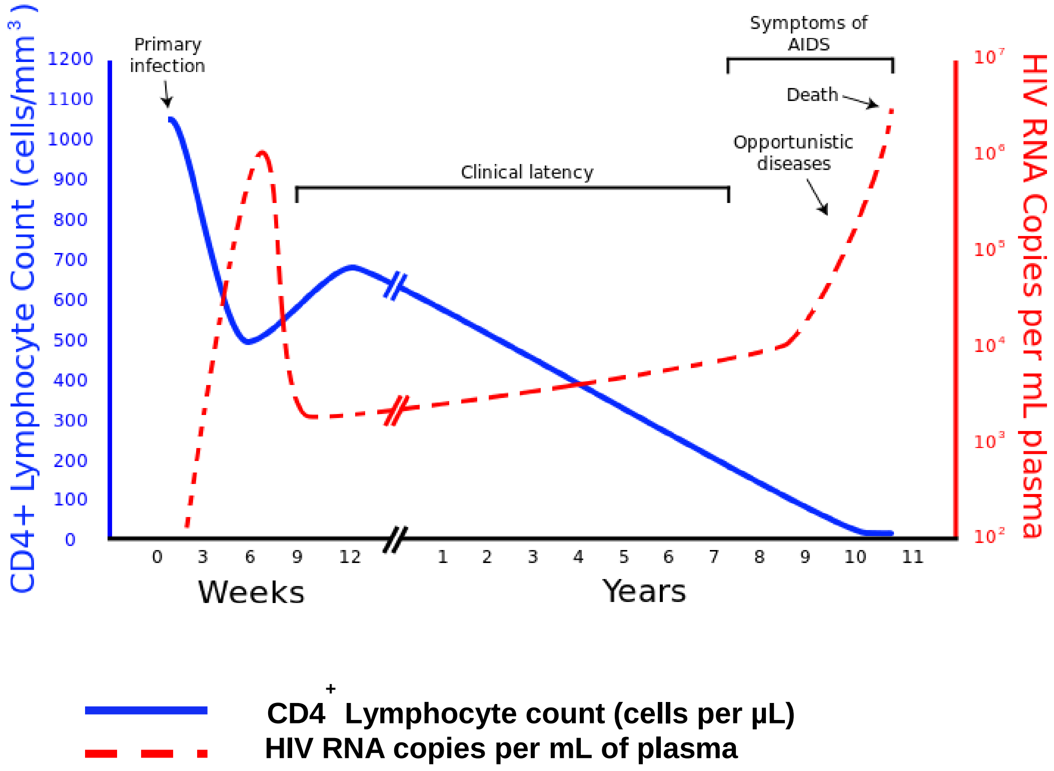


Figure 6.1

i) Describe the relationship between the graph of CD4+ Lymphocyte count and the graph of HIV RNA copies. [1]

- Inverse relationship/ as the graph of CD4+ lymphocyte count decreases, the graph of HIV RNA copies increases and vice versa.

ii) Explain the relationship stated in (a). [4]

- CD4+ lymphocytes are the specific host cells for HIV virus,
- The specific glycoproteins (gp120/gp41) on the surface of the HIV viral envelope binds to the specific CD4 receptors on the lymphocytes, facilitating entry of HIV virus
- Allows replication of its RNA genome, and reassembly to form large no of new HIV virions which bud off from host cell,pinching off host cell membrane
- Which can lead to host cell death, thus when RNA copies increases, CD4+ lymphocyte count decreases (and vice versa).

iii) As indicated in the graph above, clinical latency refers to the period where HIV infection remains undetected and patients are mostly symptom free.

Explain how the HIV virus enables itself to remain undetected during clinical latency. [2]

- Single stranded RNA genome of the HIV virus is able to be reverse transcribed to form double stranded cDNA using reverse transcriptase enzyme from HIV virus;
- Intergrase enzyme allows double stranded cDNA to be integrated with host cell genome and replicate as host genome replicates; (not recognised as foreign/ not transcribed to form viral proteins)

Therefore no clinical symptoms are caused, HIV infection is undetected.

(b) A population of *E.Coli* was grown on a culture medium containing a mixture of glucose and lactose for over 4 hours. The growth curve was plotted onto the graph in Figure 6.2 as shown.

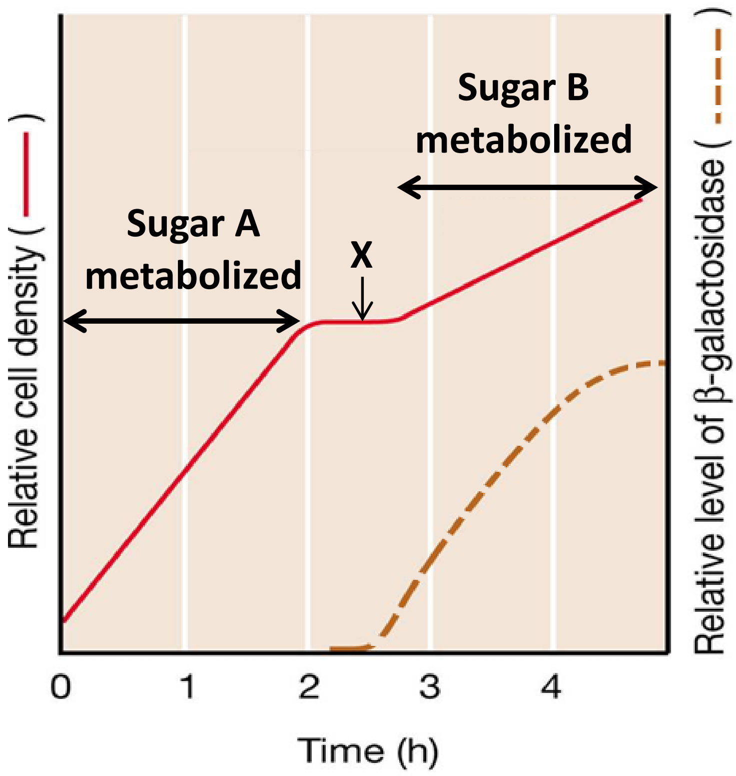


Figure 6.2

i) Identify sugars A and B. [1]

- A: glucose, B: lactose

ii) Outline the events that are taking place along the lac operon of *E coli* during the metabolism of sugar after 3 hours. [4]

- After 3 hours, concentration of glucose is very low while concentration of lactose is high, thus cAMP levels increases.
- cAMP binds to the allosteric site on CAP, alters CAP to an active conformation.

- Active CAP binds to CAP binding site on DNA, bends the DNA to facilitate binding of RNA polymerase to the promoter of the lac operon for transcription of structural genes in lac operon.
- At the same time, lactose is converted to allolactose inducer by b-galactosidase, inactivating the lac repressor. Lac repressor does not bind to operator of lac operon.

iii) Suggest an explanation for the plateau marked X in Figure 6.2. [1]

- Plateau X indicates a time delay as time is required for the lac operon to be completely induced and the production of enzymes like beta-galactosidase.

- Any possible answer accepted.

[Total: 13 marks]

Question 7

a) Green plants undergo photosynthesis and respiration simultaneously in the presence of light and oxygen. In an experiment, only the photosynthetic rate of two species of plants was examined and represented in Figure 7.1. Both plants were distinct in their leaf growth pattern. Plant A had vertical leaves while Plant B had horizontal leaves.

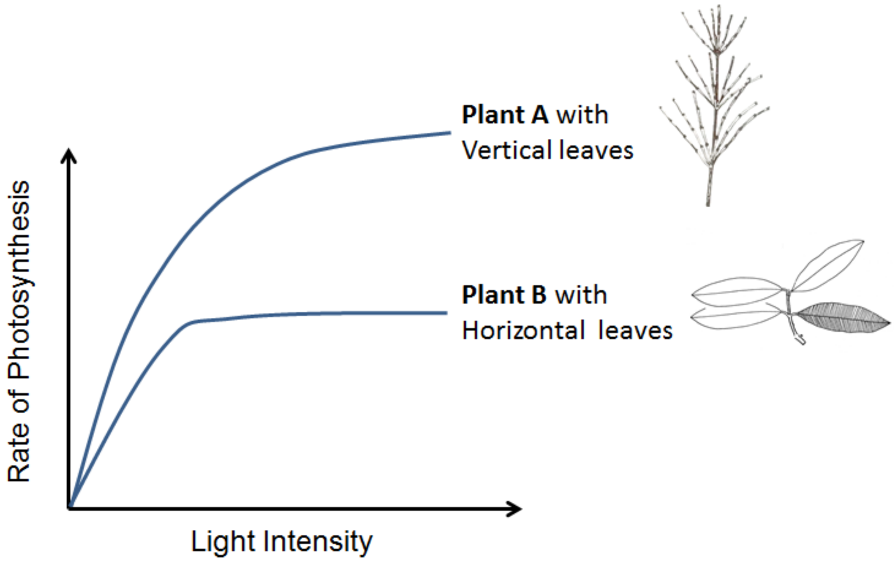


Figure 7.1

i) Explain the difference in the shape of the graphs for Plant A and Plant B in Figure 7.1. [4]

- The maximum rate of photosynthesis in Plant A is higher than that in Plant B.
- The horizontal leaves found on the top region of plant B block light from reaching the leaves at the bottom region of plant B, while the lower leaves in plant A are not blocked from light.
- Hence only the chlorophyll of the leaves in the top region of plant B reach light saturation point as compared to plant A where chlorophyll in all the leaves can reach light saturation point eventually.
- Less electrons emitted in plant B as compared to plant A, less ATP and NADPH produced, lower rate of photosynthesis as compared to plant A.

OR

- The rate of photosynthesis in plant B reaches a maximum at lower light intensity than plant A.
- The horizontal leaves found on the top region of plant B block light from reaching the leaves at the bottom region of plant B, while the lower leaves in plant A are not blocked from light.
- At lower light intensity, all the chlorophylls in plant B that are exposed to light reach light saturation point, while not all the chlorophyll in plant A have reached light saturation point.

- Thus at lower light intensity, chlorophyll in plant B emits maximum number of electrons, and produces ATP and NADPH maximally, thus reaches maximum photosynthesis rate but plant A does not.

b) Plant B was extracted and placed into another environment. Conditions of the previous and new environments are tabled below in Figure 7.2.

Condition	Quantity	
	Previous Environment	New Environment
Amount of light	80%	0%
Amount of CO_2	50%	0%
Amount of O_2	21%	21%
Water availability	Yes	Yes

Figure 7.2

i) Identify the process that is occurring within Plant B. [1]

- Aerobic respiration

ii) Explain the importance of oxygen in the identified process. [3]

- Oxygen serves as final electron acceptor at the end of the electron transport chain, accepting electrons and combining with H+ to form water via cytochrome oxidase.
- Electronegative oxygen helps to maintain the pull of electrons down the electron transport chain.
- As electrons are pulled down the ETC comprising electron carriers of progressively lower energy levels, energy released is used to maintain a proton gradient across inner mitochondrial membrane for ATP synthesis (chemiosmosis)

iii) Predict and explain briefly how the level of $NADP^+$ and NAD^+ in Plant B in this new environment compares to that in the previous environment in part a) of the question. [2]

- In the new environment, aerobic respiration is occurring, NADP+ level increases, NADP+ is only used up and converted to NADPH during photosynthesis and not aerobic respiration.
- In the new environment, NAD+ level remains similar to that in the previous environment, aerobic respiration occurs at a constant rate in both environments.

[Total: 10 marks]

Question 8

Radioactively-labelled amino acids were introduced into Cell A and Cell B. The muddle-headed researcher could not recall the identities of the cells, which was either the gastric epithelial cell or the cardiac muscle cell. The amount of radioactivity within three organelles (golgi apparatus, ribosomes, rough endoplasmic reticulum) in both cells were monitored over 45 min and tabled as shown in Figure 8.1.

	Radioactive levels / au							
	Cell A				Cell B			
	0 min	15 min	30 min	45 min	0 min	15 min	30 min	45 min
Organelle X	0	80	50	10	0	80	10	0
Organelle Y	0	0	20	15	0	60	25	5
Organelle Z	0	0	0	5	0	0	60	20

(a) Identify the organelles X, Y and Z. [1]

- X: ribosome, Y: RER. Z: GA
  (Award 1mark if all are correct)

(b) Identify the cell types of cells A and B and explain your answer. [3]

- Cell A is the cardiac muscle cell while Cell B is the gastric epithelial cell.
- Radioactivity from radioactive amino acids in cell A is mainly found in free ribosomes (organelle X) and less found in bound ribosomes on RER (organelle Y), thus cell is not a secretory cell and proteins made remain in the cell.
- Radioactivity from radioactive amino acids in cell B is passed from ribosomes in RER to the RER lumen to the Golgi apparatus, which lies on the secretory pathway of protein synthesis, thus Cell B is a secretory cell – gastric epithelial cell.

Essay

Outline the differences between prokaryotic control of gene expression with the eukaryotic model. [7]

	Point of comparison	Prokaryotes	Eukaryote
a)	Control	-At transcriptional level -No Genomic -No Post transcirptional -No Translational -No Post-translational	-Mainly at transciritonal level -Genomic -Post transcirptional -Translational -Post-translational
b)	Type of transcription factors	General transcription factors – bind to promoter	General transcription factors – bind to promoter Specific transcription factors – bind to enhancers and silencers
c)	Operon	Allows bacteria to coordinately regulate a group of genes that encode gene products with related functions.	Not present/ not significant
d)	Promoter	One promoter controlling a group of structural genes coding for enzymes involved in the same metabolic pathway	One promoter for each gene
e)	mRNA	Polycistronic mRNA one mRNA code for more than 1 protein Presence of several start and stop codons	Monocistronic mRNA one mRNA code for 1 protein 1 start codon 1 stop codon
f)	Enhancer	Absent/ not significant	Major control; increase efficiency of transcription
g)	Silencer	Absent/ not significant	Major control; decrease efficiency of transcription
h)	Accessibility of RNA polymerase to promoter	Absent/ not significant	Eg: Histone Acetylation / DNA methylation /
i	Presence of introns and therefore spliceosome	No introns no need for post transcriptional modifications so no spliceosome	Intorns => need for need for post transcriptional modifications =>spliceosome

Explain how the **molecular structure** of the axon membrane facilitates nervous transmission along the axon. [8]

- Phospholipid bilayer;
- with a hydrophobic core;
- Prevents direct entry of charged ions Na^+, K^+ into the axon;
- Allows accumulation of ions on one side of the membrane;

- Proteins distributed on the membrane;
- Leak channels for Na+ and K+
- Always open and allows ions to move down concentration gradient by *facilitated diffusion*;
- Helps maintain resting potential (with Na+/K+ pump);

- **Voltage-gated channels** for Na^+ and K^+ ions;
- Acts as channels for *facilitated diffusion* of ions down the respective concentration gradients of ions;
- Opening of Na^+ channels allows Na^+ influx and depolarization of membrane resulting in an action potential;
- Opening of K^+ channels allows K^+ efflux and repolarization of membrane;

- Na+/K+ pump;
- Actively transport ions against concentration gradient/ 3 Na+ out and 2 K+ in;
- Maintains the resting membrane potential / maintain unequal concentration gradients of Na^+ and K^+ across the membrane;
- Along with the selective permeability of membrane to K+ over Na+ ions; OR supported by more leak channels for K+ present than Na+ leak channels on the membrane;

(c) Outline the significance of post-translational modifications of eukaryotic proteins. [6]

a) Cleavage and/or covalent modification
Give 1 suitable example - glycosylation, disulfide bond formation, attachment of prosthetic groups etc. is required.

b) Form functional protein - newly synthesized proteins need to be modified for proper assembly / functioning

c) Regulate - control cellular activity / influence biological activity

d) Eg: phosphorylation/dephosphorylation may activate/inactivate (ORA) proteins

e) Degrade proteins allows control of protein activities OR prevent aberrant activities so that proteins will not stay too long in cytoplasm and still be active

f) E.g. Proteins are linked to ubiquitin that will target a protein for degradation.

g) (Save/recycle resources) - proteins not needed can be hydrolysed to amino acids, to be used for synthesis of new proteins

h) (Heterogeneity) - many different proteins modified from one polypeptide serve different function so smaller no of proteins/ small genome needed

i) (Localisation) – direct proteins to particular locations inside and outside cell

j) Eg: **modifications** at terminus of amino acid chain help target proteins for transporting to final destination in the cell / move across membranes/ tag proteins to be incorporated in various cellular and organelle membranes

Question 10

a) Compare the Calvin cycle and the Krebs cycle. [6]

Similarities

I. Involves regeneration of the initial compound, Krebs cycle - oxaloacetate & Calvin cycle - ribulose bisphosphate (RuBP);

II. Involve oxidation-reduction / redox reactions;

III. Involve co-enzymes as electron and proton carriers, Krebs cycle – NAD^+ & FAD, In Calvin cycle – reduced NADP / NADPH.

Differences

Features	**Calvin cycle**	**Krebs cycle**
Site	stroma of chloroplast	matrix of mitochondria
Nature of process	Anabolic - formation of triose phosphate or starch	Catabolic - breakdown of acetyl coA
Role of carbon dioxide	1 molecule of CO_2 is accepted by RuBP catalyzed by RuBP carboxylase - oxygenase	2 molecules of CO_2 is released by oxidative decarboxylation
Coenzymes involved	Reduced NADP / NADPH + H^+ as electron and proton donor	NAD^+ & FAD as electron and proton acceptors
Role of ATP	Expenditure of ATP - in the reduction of PGA / GP to TP / GALP and in regeneration of RuBP	Synthesis of ATP by substrate level phosphorylation

b) Explain the importance of operons in bacteria. [4]

1. Small genome
2. need to pack all essential genes;
3. save space;
4. Operons are a linear arrangement of genes and its regulatory sites;
5. arrangement allow genes that encode proteins that execute similar functions to cluster together;
6. allows for proteins performing similar functions to be turn on together;
7. Allow for common mode of regulation;
   Able to share common promoter;

c) Describe the processes through which genetic diversity can be achieved in bacteria. [10]
   mutation

   Transformation (max 3)

   1. Uptake of naked foreign DNA from surrounding external environment into recipient bacteria cell
   2. DNA binds to cell surface, facilitated by DNA binding proteins
   3. Nuclease degrades one strand of double stranded foreign donor DNA, single stranded DNA enters
   4. Single stranded DNA is then integrated into the bacterial chromosome by means of RecA proteins
   5. Recombination between homologous regions
   6. Can be either artificial or natural process
   7. Cells must be competent.

   Transduction (max 4)

   8. Process by which DNA is transferred from donor bacterial cell to another recipient by bacteriophages
   9. Generalised and Specialised transduction both result from aberrations in the phage reproductive cycle
   10. Generalised transduction: random segment of host bacterial chromosome is transferred to recipient bacterial cell
   11. During assembly stage, a small piece of host cell's degraded DNA is packaged within the phage capsid in place of the phage genome.
   12. Transducing phage infects recipient cell, injects the DNA acquired from donor bacterium into recipient bacterium
   13. Donor DNA is integrated into recipient bacterial chromosome by homologous recombination.
   14. Specialised transduction: occurs in temperate bacteriophages, involves transfer of a specific set of bacterial genes from donor to recipient bacterium.
   15. In the lysogenic cycle, the genome of a temperate phage integrates as a prophage into the host bacterium's chromosome at a specific site.
   16. When the phage genome is excised from the host bacterium's chromosomes, sometimes a small region of the host bacterial DNA adjacent to the prophage is excised together.
   17. The phage DNA that incorporated some bacterial genes replicates and is packaged to form a specialized transducing phage.

18. Specialized transducing phage released during host cell lysis to infect a recipient bacterial cell.
19. Foreign bacterial DNA subsequently integrated into the recipient bacterial chromosome by homologous recombination.

Conjugation (max 3)

20. Transfer of F plasmid from F+ donor to F- recipient cell
21. Via sex pilus / conjugation tube encoded by genes in F plasmid
22. Sex pilus attaches to F- cell and retracts, forming cytoplasmic bridge
23. Nuclease cleaves one strand of double stranded F plasmid at Ori, single stranded F DNA of F plasmid transferred
24. Rolling circle replication
25. Both cells end up with F plasmid and are thus both F+.